

Manufacturing process Week-5

1. Explain forging Operations -Upsetting, drawing down, Cutting, Bending
2. Explain Punching and Drifting, Setting down and Finishing, Forge Welding
- 3 Estimation of Length of Raw material required to convert Circular rod to Square and Calculate the Production Cost considering direct and Indirect expenses.

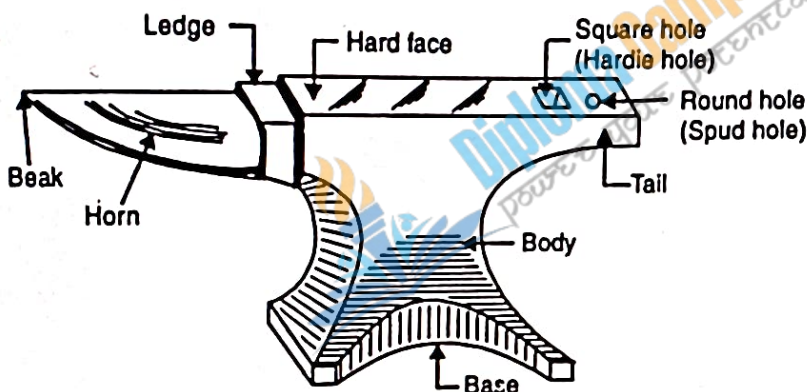
Forging

Forging is a process in which the metal is plastically deformed to the desired shape, by applying pressure with the help of pair of dies. One die is stationary and the other has a linear motion. Forging process can be carried out both in the hot and cold states of the metal.

FORGING TOOLS

1. Anvil:

The Anvil block gives the rigid support during forging. The anvil must be of such a weight that it can absorb the blows that are struck upon it. The body is made of mild steel. The commonly used size of an anvil weighs approximately 50-150 kg although it is manufactured in various sizes.



2. Hammer:

The hammer has a heavy head fixed with a long handle which gives a high impact load on the small area of the workpiece. It is used for striking the hot or cold workpiece to change shape and size as needed.



4. Chisels:

Chisels are the tools used for cutting metal.

Two types of Chisels (1) Hot Chisel (2) Cold chisel.

The cold chisel is used for cutting cold metals while the hot chisel is for hot metals.

Cold chisel is made of high carbon steel.

A hot chisel is made of medium carbon steel as there is no need of hardening. A hot chisel is used to cut the metal in plastic state.



5. Punches:

Punches are tapered tools made in various shapes and sizes. They are used for producing holes in red hot jobs. A larger tapered punch is called a drift.



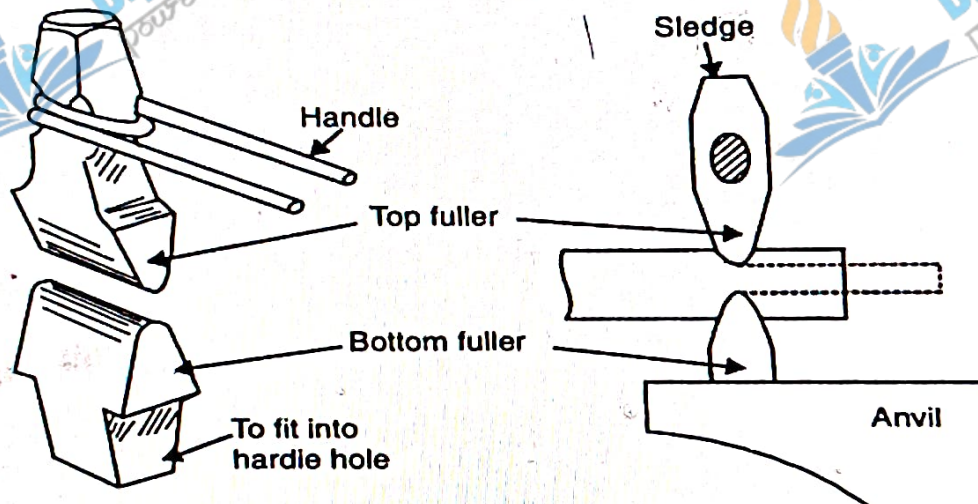
Flatters:

Flatters are made of high carbon steel and consist of a square body, fitted with a handle, and a flat square bottom. They are used for leveling and finishing a flat surface after drawing out or any other forging operation.



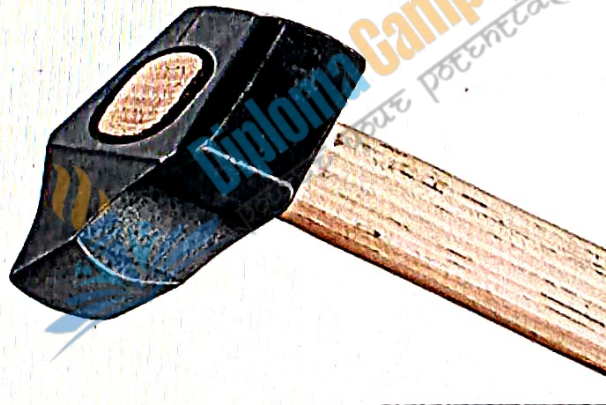
Fullers:

Fullers are made of high carbon steel in different sizes to suit the various types of jobs. They are usually used in pairs, consisting of a top and bottom filler. Their working edges are normally rounded. They are employed for making necks by reducing the cross-section of a job and also in drawing out.



(a) Top and bottom fullers.

(b) Use of fullers.



Swages:

Swages are made of high carbon steel in two parts called the top and bottom swages.

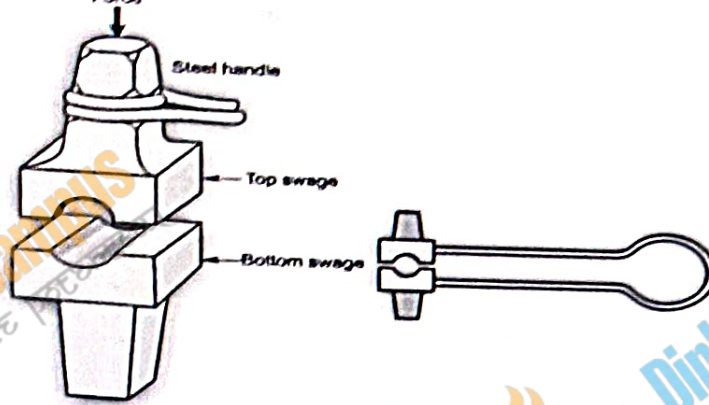
They are available in various sizes.

They are used for increasing the length of a circular rod or for finishing the circular surface of a job after forging.

Force
↓



Visual Watermark



(a) A top and bottom swage

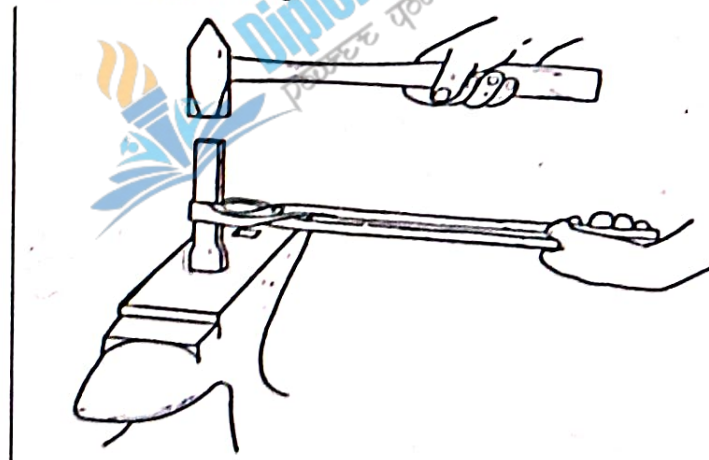
(b) A spring swage

The forging operations that are usually required in shaping the part are:

1. Upsetting
2. Drawing down
3. Bending
4. Edging or Rolling
5. Fullering or Swagging
6. Cutting out
7. Punching and Drifting
8. Forge welding
9. Swaging
10. Flattening and setting down

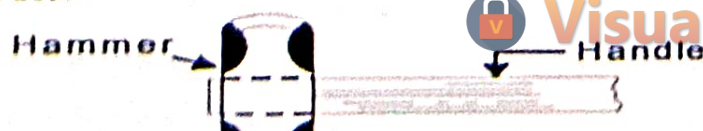
1. Upsetting

Upsetting is forging operation carried out to increase the thickness of a section at any desired portion at the expense of decrease in length section.



2. Drawing down

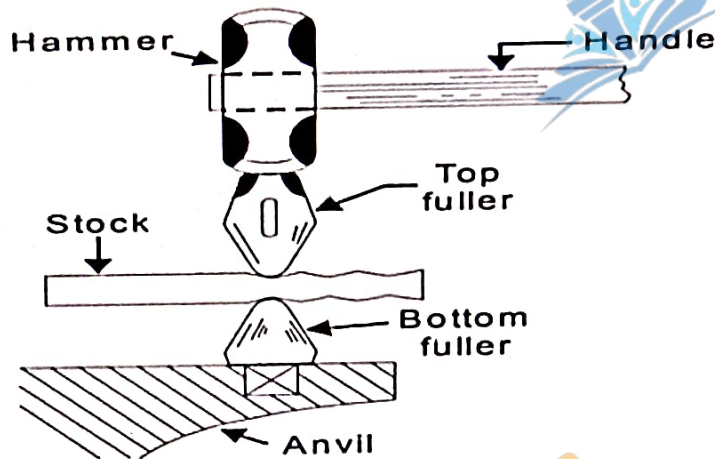
It is the process of increasing the length of any piece and at the same time reducing either its width or thickness or both.



Visual Watermark

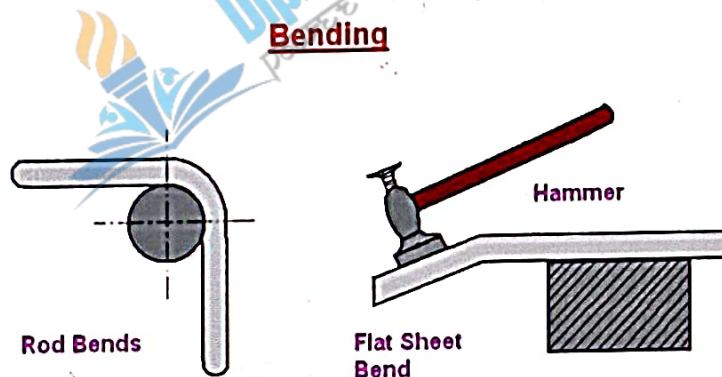
2.Drawing down

It is the process of increasing the length of any piece and at the same time reducing either its width or thickness or both.



3.Bending

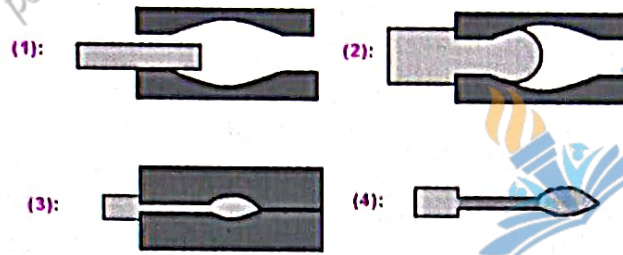
Bending is the act of turning or bending a piece of metal in a round or angular shape. This process is used to create round rings and other shapes. Bending operation is the most common and easiest forging. It is done to give a turn shape to a metal rod or plate.



4.Edging

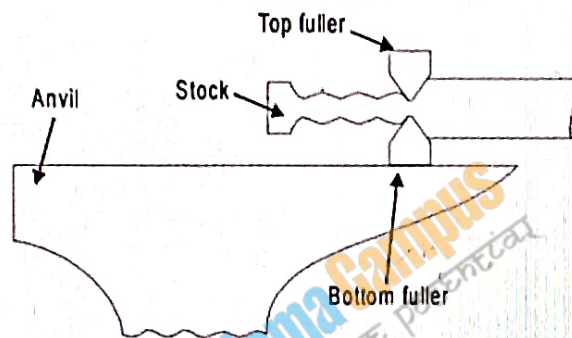
The process (edging or rolling operation) of distribute the metal longitudinally by moving the metal from the portion of the stock where it is in excess to the portion which is deficient in metal.

Edging



5.Fullering

It is the operation of reducing the stock between the two ends of the stock at a central place, so as to increase its length.

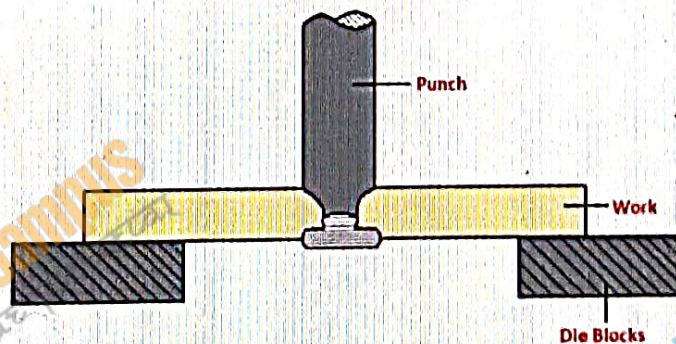


6.Cutting out

This operation consists of workpiece being cut completely in trimming to shape. The long piece is cut into several specified lengths. This operation is performed with the help of chisels and hardy.

7.Punching and drifting

Punching is a process by which through holes or recesses are made in the workpieces when it is heated to plastic state. These recesses may be parallel or tapered and may be of any shape and size. Drifting is the process of widening and smoothening of the punched hole by passing a tapered drift of the exact size of the recess.



Punching Operation

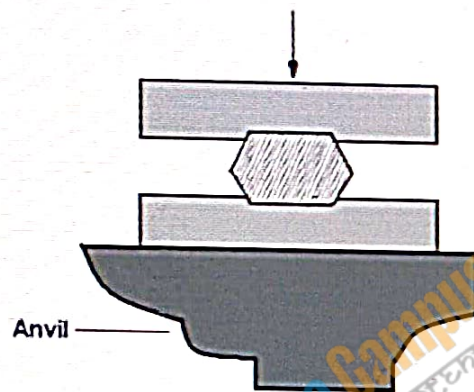


8. Forge Welding

It is the process by which the separately forged pieces or parts are joined together. Workpieces to be joined are heated to white heat, beyond the plastic range but below the fusion point. Further they are joined them into one solid piece by forcing them one into another with hammer blows. The common examples of forge welding are hooks in chains, ends of rings, links.

9. Swaging

Swaging is done to reduce and finish work for desire size and shape, usually either round or hexagonal. For small jobs top and bottom swage pair is employed, where as for large work swage block can be used.

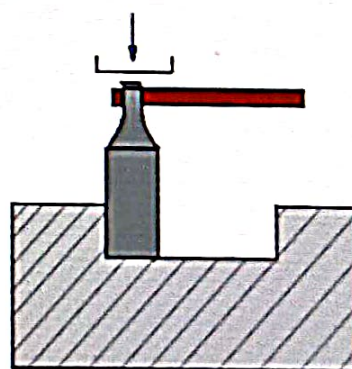
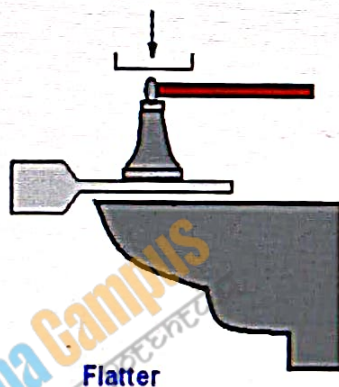


Swaging

10. Flatting and setting down

Flatting and setting down are done to obtain a smooth surface with the use of a flatter or set hammer. This operation is done after fullering leaves a corrugated surface of the metal, or after the workpiece is forged into shape with a hammer.

Flatting and Setting Down



Set Hammer

FORGING SQUARE SHAPED BAR

Aim

To forge a square shaped bar from a given cylindrical rod

Tools Required

Steel rule, caliper, anvil, sledge hammer, flat tong, flatter, etc

Procedure

- The final length of the model to be forged is calculated suitably. Refer figure 1.24
- Ignite the coal in the furnace. Switch on the blower and charge sufficient coal into the furnace
- The given cylindrical shaped work piece is placed in the hearth and heated to red hot temperature.
- Draw down the heated work piece to the calculated length with the help of hammer, tongs and anvil. The process is carried out appropriately until the circular rod is transformed into square shape of the desired dimensions
- With the help of flatter, finish the work piece to the final dimensions and surface finish
- The specimen is allowed to cool in air.

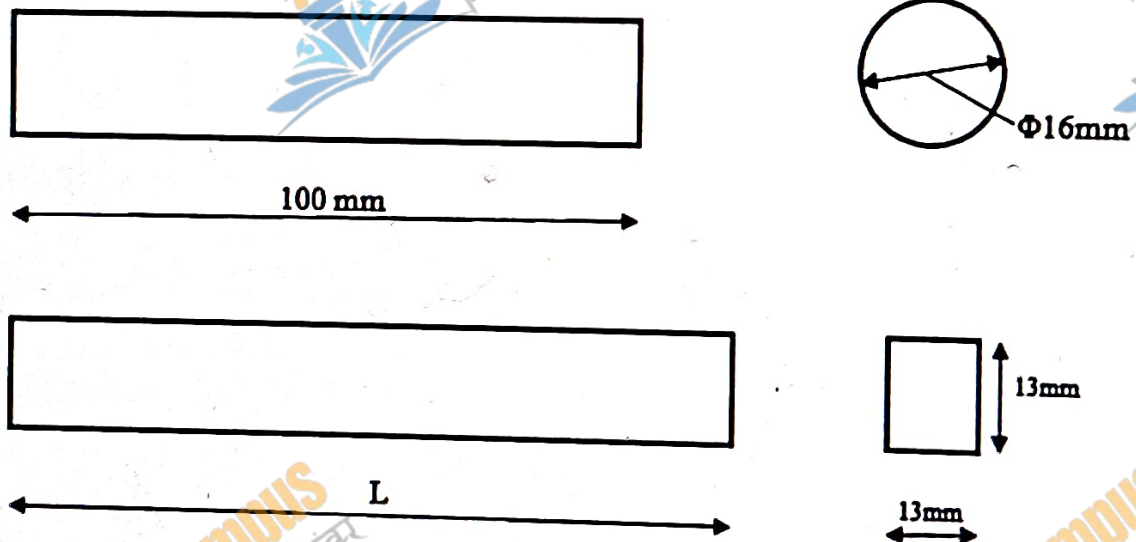


Fig. 1.24 Forging Square Shaped Bar



Specimen calculation

Volume of given material = Volume of finished model

or

Cross-sectional area x length = cross-sectional area x length

$$\frac{\pi}{4} D^2 l = L \times b \times h$$

$$\frac{\pi}{4} \times 16^2 \times 100 = L \times 13 \times 13$$

Length of square rod = L = 119 mm

Compensating 6 % wastage for loss in material during forging, we get

$$\text{Wastage} = \frac{6}{100} \times 119$$

Wastage = 7 mm

Final length of square specimen = L = 119 - 7 = 112 mm

Thus, if the specimen is drawn suitably to a length of 112 mm, a circular rod of 16 mm can be shaped to a square rod of sides 13 mm.

Result

The given circular rod is forged to a square shaped model.